

Pierre **Courteille**

Cloud Architect · DevOps Engineer

Geneva, Switzerland

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Education

University of Montpellier — *Montpellier, France*

Master's, Mathematics & Computer Science (Master MOCA — modeling, optimization, combinatorics, algorithms) — *2012–2015*

University of Nantes — *Nantes, France*

Bachelor's, Mathematics & Computer Science — *2009–2012*

Skills

DevOps & platforms	AWS (Organizations, CloudFormation, CDK, CloudWatch, Systems Manager), Docker, Kubernetes (EKS), ECS Fargate, GitLab CI/CD, CodePipeline, CodeBuild, Jenkins, Terraform
Cloud architecture	Well-Architected pillars, multi-account landing zones, microservices, event-driven design, API-first and serverless (Lambda, API Gateway), multi-AZ design, high availability, multi-region and disaster recovery
Networking & connectivity	VPC (subnets, route tables, NACLs, security groups), Transit Gateway (hub-and-spoke, VPC/VPN/Direct Connect attachments, inter-region peering), hybrid connectivity (Site-to-Site VPN, Direct Connect patterns), Route 53, VPC endpoints, PrivateLink
Security & governance	IAM, IAM Identity Center, SCPs, least privilege, KMS, WAF, Security Hub, AWS Config, VPC endpoints, secure cross-account access
Data & integration	RDS, DynamoDB, Redshift, ElastiCache, S3, messaging and integration (SQS, SNS, EventBridge), log/query (Athena, CloudWatch Logs Insights)
Observability & FinOps	CloudWatch OAM, Grafana, X-Ray, centralized logging/metrics/alerting, budgets, Cost Explorer, CUR-based reporting
Programming & automation	Python (incl. Boto3), C#, Node.js, JAVA, C/C++, .NET Core, shell scripting, infrastructure automation (CDK, Terraform, CloudFormation, AWS SAM)
Management, mentoring & collaboration	Team coordination and delivery alignment; mentoring and knowledge transfer (documentation, workshops, pair sessions); cross-functional collaboration with product, security, and business stakeholders; facilitation of architecture reviews, onboarding, and operational handovers
Languages	French, English, Spanish

Certificates

Jul 2025 **AWS Certified DevOps Engineer – Professional (DOP-C02),** Amazon Web Services (AWS)
Jul 2025 **AWS Certified Solutions Architect – Associate (SAA-C03),** Amazon Web Services (AWS)
Jul 2025 **AWS Certified Developer – Associate (DVA-C02),** Amazon Web Services (AWS)
Jun 2025 **AWS Certified Cloud Practitioner (CLF-C02),** Amazon Web Services (AWS)

Work Experience

Qim info / International Electrotechnical Commission (IEC)

Geneva, Switzerland

Cloud Engineer

Oct. 2025 - Present

The IEC is a Geneva-based international standards organization for electrical, electronic, and related technologies. Within IT, I own AWS topics end to end: design and delivery of a secure, scalable multi-tenant SaaS platform, plus continuous improvement of monitoring, observability, networking, and security to raise reliability, visibility, and performance.

- **Networking:** multi-AZ VPCs with public, private, and isolated subnets; dedicated VPC per tenant with strict Security Group policies; controlled interconnectivity with AWS Transit Gateway.
- **Compute & orchestration:** containerized workloads on ECS (Fargate); event-driven patterns with Lambda and API Gateway; EC2 with Auto Scaling Groups and ALB/NLB where workloads require it.
- **Data & caching:** RDS (Aurora MySQL/PostgreSQL, Aurora Serverless v2); DocumentDB for document-oriented needs; ElastiCache (Redis, Memcached) for application caching and latency.
- **IaC & tenant delivery:** one modular, reusable AWS CDK codebase to deploy full per-tenant infrastructure automatically; CloudFormation and Terraform for legacy or specific stacks; standardized architecture patterns and reproducible environments; industrialized client onboarding (AWS accounts + full stack provisioned in minutes).
- **Observability (multi-account / multi-region):** centralized observability in a dedicated monitoring account under AWS Organizations; CloudWatch Observability Access Manager (OAM); centralized metrics, logs, and traces; Amazon Managed Grafana; Prometheus-style application metrics; alerting with CloudWatch Alarms and SNS.
- **CI/CD & automation:** advanced GitLab CI/CD pipelines for secure, automated infrastructure and application rollout across AWS accounts; dynamic, parameterized pipelines; idempotent deployments; build, test, and deploy microservices; IAM roles via STS (AssumeRole); CI tuning (parallel jobs, shorter build/deploy cycles).

Stack: AWS (Organizations, CDK, CloudFormation, Terraform, ECS Fargate, Lambda, API Gateway, RDS, Aurora, DocumentDB, ElastiCache, Transit Gateway, CloudWatch, OAM, Grafana, GitLab CI, SNS, EC2, ALB, NLB), Python.

Premaccess

Paris, France

Cloud Engineer (freelance)

Jun. 2019 - Sep. 2025

Premaccess is a Franco-Swiss IT managed services company focused on digital transformation and AWS. I supported many customers—including Smartpush, ATA Electronics, Cap Adrenaline, BNC, Fitizzy, Truffaut, Prestigia, and others—through full cloud migrations, ongoing evolution, and continuous optimization of their AWS estates.

- **Cloud adoption & growth:** end-to-end migrations to AWS, then iterative modernization as needs grew (containers, serverless, data platforms, edge distribution).
- **Network & security:** multi-AZ VPCs (public / private / isolated subnets); hardened Security Groups and NACLs; VPC Endpoints and VPC Flow Logs; centralized identity with IAM Identity Center, IAM, and SCPs; encryption with KMS.
- **Compute & orchestration:** ECS (Fargate) and EKS for resilient microservices; Lambda + API Gateway for event-driven workloads; EC2 with Auto Scaling and ALB/NLB when required.
- **Data & analytics:** RDS (Aurora MySQL/PostgreSQL, Serverless v2); DynamoDB for low-latency NoSQL; Redshift for warehousing and heavy analytics; ElastiCache where applicable.
- **Messaging & integration:** SQS, SNS, and EventBridge for asynchronous and event-driven integration.
- **Edge & distribution:** CloudFront with OAC/OAI and WAF; Route 53 and ACM for DNS and TLS automation.
- **Observability, security ops & FinOps:** CloudWatch Logs/Metrics/Alarms, X-Ray, GuardDuty, AWS Config, Security Hub; cost dashboards (Cost Explorer, Budgets, CUR) and FinOps-oriented optimization.
- **CI/CD & IaC:** AWS CDK, CloudFormation, and Terraform; CodePipeline, CodeBuild, CodeDeploy, GitLab CI, and Jenkins for automated test, build, and release.
- **Engineering deliveries:** designed and built internal BAM tooling (Python, Boto3) for one-click provisioning and teardown of ephemeral environments; C# refactor delivering ∼30% lower response time and ∼40% higher TPS; AWS SAM for industrialized Lambda and REST APIs with DynamoDB and CloudWatch Logs; L2/L3 support, security-oriented reviews, and architecture assessments.

Stack: AWS (EC2, ASG, ALB/NLB, ECS, EKS, Organizations, IAM Identity Center, Lambda, RDS, OpenSearch, DynamoDB, ElastiCache, SQS, SNS, EventBridge, CloudFront, S3, KMS, WAF, Route 53, CloudWatch), Python, Node.js, .NET Core, CDK, CloudFormation, Terraform, GitLab, Jenkins, CodePipeline, Docker, Kubernetes.

ISIKA

Paris, France

.NET Instructor & Software Engineer (freelance)

Mar. 2020 - Apr. 2023

C# and .NET training with a focus on OOP and application design. Combined formal teaching with hands-on engineering so learners could ship small end-to-end applications.

- **Technical training:** C# fundamentals, OOP (classes, inheritance, polymorphism), desktop development (WinForms), debugging and code structure.
- **Application development:** guided learners through full apps including data persistence (LINQ, Entity Framework, MySQL), layering, and basic testing mindset.
- **Pedagogy:** project follow-up, individual coaching, rubrics for assessed deliverables; adapted pace to professional audiences.
- **Course material:** maintained exercises and examples aligned with workplace scenarios (CRUD, validation, simple UI patterns).

Stack: C#, HTML5, Entity Framework, LINQ, MySQL, WinForms.

Bassetti

Kolkata area, India

Software Quality Assurance Manager

Feb. 2016 - May 2018

Built and led a QA organization in India: team leadership, large-scale test automation, and DevOps practices. Focus on throughput, traceability, and predictable releases.

- **Hiring & leadership:** recruited, trained, and led 25+ QA engineers; career paths, skills matrix, and on-boarding for distributed teams.
- **Automation & tooling:** co-designed an internal platform executing 10,000+ API tests/hour; improved coverage, flakiness reduction, and reporting for leadership.
- **Quality & delivery:** tight collaboration with development; automated feedback loops, daily test runs, shorter regression cycles before releases.
- **Process & DevOps:** introduced DevOps-oriented practices (shared ownership of quality gates, CI integration), clearer QA–dev handoffs, faster cadence without sacrificing critical checks.
- **Metrics:** tracked defect trends, escape rate, and automation ROI to steer investment in the right layers of the stack.

Stack: C#, .NET, Selenium, Jenkins, APIs, XML, Excel.

Kronos Incorporated (Ad Opt)

Montreal, Canada

Optimization Engineer

Jan. 2015 - Jan. 2016

Ad Opt (Kronos) — optimization for airline crew planning (cost and operational efficiency). Mixed R&D, heavy computational experiments, and customer-facing delivery.

- **Algorithmic optimization:** performance gains via column generation and shortest-path strategies on large instances; profiling and memory-aware implementations in C/C++.
- **Validation:** intensive testing on thousands of instances; stress cases for edge constraints and stability before production rollout.
- **Production:** customer deployments, integration support, and tuning with operations teams.
- **R&D:** presentations at GERAD; contributions to scientific publications; knowledge sharing on modeling choices and solver behavior.
- **Engineering practice:** Linux toolchain, GDB, Git workflows; collaboration with researchers and product on requirement formalization.

Stack: C, C++, Linux, GDB, Git.